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Java Program: Bank Account with Custom Exception

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// Name: Harshwardhan Malojirao Galande
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// Java program to handle deposit and withdrawal using custom exception

class InsufficientFundsException extends Exception {
    public InsufficientFundsException(String message) {
        super(message);
    }
}

class BankAccount {
    private double balance;

    public BankAccount(double initialBalance) {
        this.balance = initialBalance;
    }

    public void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited: " + amount);
    }

    public void withdraw(double amount) throws InsufficientFundsException {
        if (amount > balance) {
            throw new InsufficientFundsException("Insufficient balance! Cannot withdraw " + amount);
        }
        balance -= amount;
        System.out.println("Withdrawn: " + amount);
    }

    public double getBalance() {
        return balance;
    }
}

public class BankApp {
    public static void main(String[] args) {
        BankAccount account = new BankAccount(1000.0); // initial balance

        try {
            account.deposit(500.0);
            account.withdraw(300.0);
            account.withdraw(1500.0); // This will throw the custom exception
        } catch (InsufficientFundsException e) {
            System.out.println("Exception: " + e.getMessage());
        } catch (Exception e) {
            System.out.println("Runtime exception occurred: " + e.getMessage());
        }

        System.out.println("Final Balance: " + account.getBalance());
    }
}
```